Learning a sequence model from noisy data alone

What the estimator buys, what it is made of, and where it stops — 25 September 2026

1. What is being measured

The setting. A diffusion model generates data by learning to undo noise, and the object it must learn at each noise level is the *score*. On real data the true score is unknown, so nobody can measure how far a learned one is from it. We work in a case where the true score is computable exactly, and ask one question: **is an estimator that knows the data has a sequential structure better than a general network that does not, when both are shown the same information?**

The data. A Markov chain: each site is a fraction of the previous one plus fresh randomness,

$$a_{i+1} \;=\; \rho\, a_i + \varepsilon_i, \qquad i = 1, \ldots, L-1.$$

ρ is the **correlation**: how strongly each site remembers the one before it. The innovation ε is drawn from a density proportional to $e^{-|x/s|^\beta}$, and the exponent β is the **shape**. The scale s is always reset so the variance of ε is the same for every β: β therefore changes how the randomness is *distributed*, never how large it is. This is what separates "non-Gaussian structure" from "more noise", and it is the distinction the whole result turns on.

β	the innovation is	excess kurtosis
1	Laplace: occasional large jumps	+3.00
1.5	slightly heavier than Gaussian	+0.76
2	Gaussian	0.00
3	lighter: jumps suppressed	−0.58

Table 1: The shape β at fixed variance. Excess kurtosis is zero exactly at the Gaussian. Our chain uses $\beta = 1$.

	is	values used
M	how many sequences we learn from	32–2048
L	how many sites in each sequence	16–128
C	flexibility of the fitted noise	1–16
t	the noise level	12 levels

Table 2: M and L describe the data; C is a dial on our estimator, not a property of the data. The 12 levels are log-spaced over $t \in [0.05, 3]$.

Our estimator, and its one dial. We fit the chain by EM: alternate between inferring what the clean sequence probably was (exactly, by message passing) and re-estimating the parameters. The innovation law is *not* assumed — it is fitted as a mixture of C Gaussians with free weights, means and variances, which can represent heavy tails, skew and bimodality alike. It is never told the truth is Laplace, and at $C = 16$ it recovers an excess kurtosis of 2.91 against the true 3.00. Setting $C = 1$ forces the innovation Gaussian; that restriction is the control in §3.

The competitors. Six neural sequence denoisers: a fully connected network, a convolutional window head, a dilated convolutional network (TCN), a bidirectional GRU, a transformer with relative-position attention, and a 1-D U-Net. Each is screened and tuned *separately inside every regime*, and the best of the six in each cell is what we report.

The rule of the comparison. Both sides see **one noisy view of each sequence and nothing else** — never the clean sequence, never a second view. Neither side may use the true score to decide when to stop training: the networks stop on a held-out noisy objective, our estimator on held-out evidence. This is the condition comparisons of this kind most often violate, and it is the one that makes the numbers below a fair fight.

The measure. Every number in this document is a **relative score error**,

$$\sqrt{\sum \|\hat{s} - s_{\text{true}}\|^2 \;\Big/\; \sum \|s_{\text{true}}\|^2}$$

summed over the twelve noise levels and the test sequences: the size of your error divided by the size of the thing you are estimating. It is dimensionless and has three landmarks. The exact score scores 0. Ignoring the sequence structure entirely scores 0.670. And 0.168 is the best score reachable by *any* method that uses correlations but nothing beyond them — a proven floor, not an empirical best, which is what lets us say what a method has learned rather than only which one won.

2. The result: structure is worth 5–10× at equal information

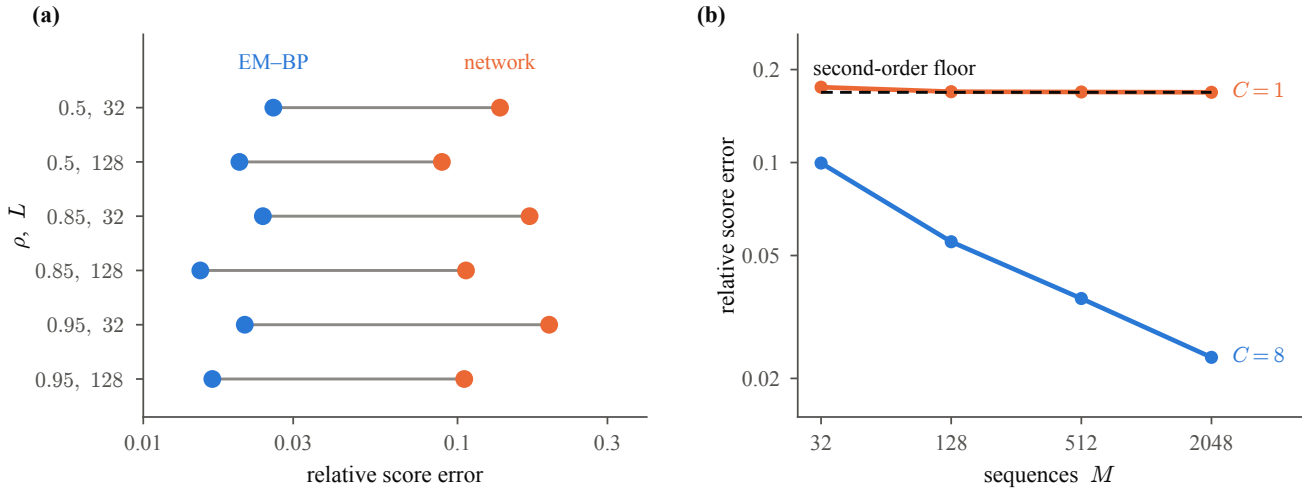


Figure 1: **(a)** Relative score error, lower is better, at six combinations of correlation ρ and length L, at $M = 2048$. Blue is our estimator, orange the best of the six networks on the same single noisy view. **(b)** The control. The dashed line is the second-order floor. The same estimator restricted to a Gaussian innovation ($C = 1$, orange) sits on that floor at every sample size and never leaves it; allowed a non-Gaussian innovation ($C = 8$, blue) it goes far below.

ρ	L	LMMSE	EM–BP (ev.)	EM–BP (bp)	best net (C)	best net (B)	factor C / EM–BP (ev.)
0.5	32	0.265	0.026	0.024	0.137	0.129	5.7 [4.9, 6.7]
0.5	128	0.268	0.020	0.018	0.089	0.079	4.6 [4.1, 5.0]
0.85	32	0.167	0.024	0.021	0.170	0.156	8.3 [6.6, 10.3]
0.85	128	0.168	0.015	0.015	0.106	0.095	8.0 [6.7, 9.2]
0.95	32	0.086	0.021	0.019	0.196	0.128	10.3 [8.8, 11.9]†6
0.95	128	0.087	0.017	0.016	0.105	0.094	6.4 [6.0, 6.9]

Table 3: 16 seeds per cell, $M = 2048$. LMMSE is the second-order floor for that regime. *EM–BP (ev.)* is our estimator stopped on held-out evidence, which is the honest rule; *(bp)* is the same estimator stopped by an oracle, shown so the cost of honesty is visible. *best net (C)* is the best network on one noisy view — the matched comparison. *best net (B)* is the best network additionally given the *clean* sequence, an easier problem included to show what the restriction costs. The factor compares net (C) with EM–BP (ev.), paired by seed, with a percentile bootstrap. †6: in 6 of the 16 seeds of that cell the network's held-out curve never improved on its first checkpoint — reported, deliberately not excluded.

How to read it. Our estimator is ahead in all six regimes and in every one of the sixteen repetitions of each, by factors from 4.6 to 10.3. Three things in the table matter beyond that headline. First, *ev.* and *bp* are nearly identical, so being denied the true score when choosing where to stop costs us almost nothing — the two rules pick the same iteration in 52 of 96 cells. Second, the conservatism runs against us: our estimator is still improving when its iteration budget runs out in 56 of 96 cells, so its error there is an upper bound. Third, the floor itself moves with ρ, and where the network sits relative to it is the real story: at $\rho = 0.95$ the network (0.196) is *worse* than the floor (0.086), meaning one noisy view per sequence is not enough for it to learn even the correlation, while at $\rho = 0.5$ it is below the floor and has genuinely learned beyond second order.

Table 4: Schedule-pooled relative score error on all sites, mean over 16 seeds (s.e. in the text macros; the isotropic reference is 0.669 at every size). Every arm is scored on identical test bundles. The last column is the same capacity fitted to the innovation law itself—the C-component mixture closest in Kullback–Leibler divergence, an infinite-data reference for the family rather than a bound; † marks fits whose narrowest component is under two grid cells.

	sequences M				
kernel	32	128	512	2048	$M = \infty$
LMMSE floor	0.167	0.167	0.167	0.167	0.167
$C = 1$	0.174	0.168	0.168	0.167	0.167
$C = 2$	0.084	0.050	0.035	0.028	0.039
$C = 4$	0.093	0.053	0.037	0.025	0.0057
$C = 8$	0.099	0.055	0.036	0.023	0.0022†
$C = 16$	0.100	0.056	0.036	0.022	0.0007†

3. The control, and the boundary

What the margin is made of. Panel (b) and the table above are a comparison of our estimator with *itself*, one dial moved. Restricted to a Gaussian innovation it lands on the second-order floor and stays there however much data it is given; allowed a non-Gaussian innovation it goes far below. Since nothing else differs, the margin is fourth-order structure and cannot be anything else. The same holds across fifteen further combinations of (ρ, L), with the restricted arm matching the theoretical floor to 0.0099 in the worst of them — which incidentally tests the grid, the message passing, the EM and the scoring at once. It also holds with the non-Gaussianity reversed, on a chain *less* heavy-tailed than Gaussian, where the factor is 8.8 [7.4, 10.2].

Two things to notice in it. At $M = 32$ the richest kernel (0.100) is *worse* than the simplest non-Gaussian one (0.084): flexibility costs variance when data is short, and the ordering only reverses by $M = 2048$. And the last column is the mixture closest in Kullback–Leibler divergence to the innovation law itself, which is a reference for how expressive the family is — not a bound on score error, and not comparable with the finite-M columns.

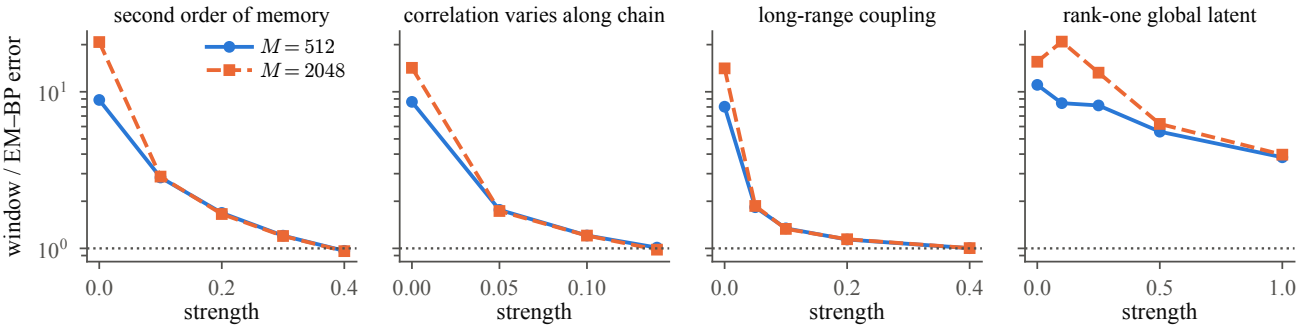


Figure 2: How wrong the chain assumption may be before the advantage goes. The vertical axis is a *ratio*: a convolutional window head's error divided by ours, so the dotted line at 1 is a tie and above it we still win. Strength 0 is the assumed model in every panel; the four departures are, left to right, a second order of memory ($a_i = \rho_1 a_{i-1} + \rho_2 a_{i-2} + \varepsilon$, strength $= \rho_2$), a correlation that drifts along the sequence ($\rho_i = \rho + \delta \cos(2\pi i/L)$, strength $= \delta$), a long-range coupling, and one shared hidden factor. All four perturbed models are Gaussian, so the true score of the *perturbed* model is computable exactly and both sides are scored against it — we are not grading ourselves against our own assumption.

Where it ends. Two departures reverse the ordering inside the range swept: a second order of memory between 0.3 and 0.4, and a drifting correlation between 0.1 and 0.14. Two never reverse it — a shared hidden factor is absorbed into the fitted correlation, and a long-range coupling decays to a tie. The limitation to state plainly: all four perturbations are Gaussian, because that is what makes the perturbed truth's score exact, so this locates the boundary against departures in the *dependency structure* and not against departures in the shape of the noise.

What the numbers rest on. Every experiment is frozen with hash-pinned outputs and a configuration registered before it ran. Each number is generated from those outputs by one tool and recomputed independently by a second that fails if the two disagree; the figures check themselves against the manuscript's numbers and refuse to be written on a mismatch. One correction is on the record: the capacity sweep in the earlier archival study is void — its training draw left every configuration fitted to a single repeated sequence — and the capacity evidence above was computed independently of it.